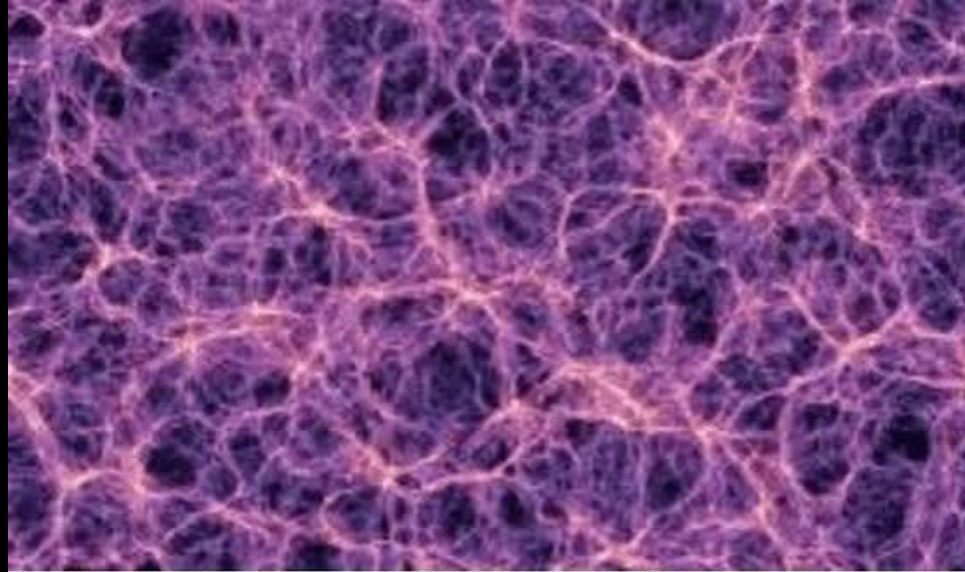


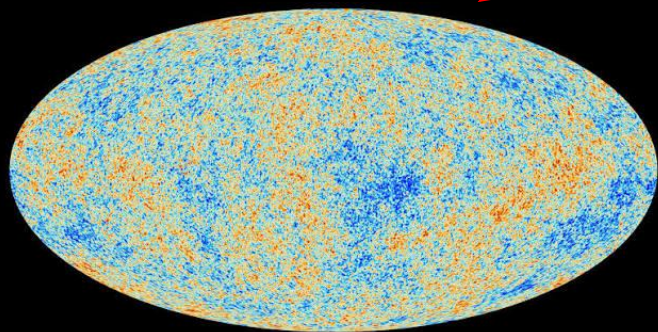
An Excursion into Non-Gaussianity



Licia Verde ICREA & ICCUB

Back in the last millennium.... (basically before recombination)

Say inflation generated some primordial non-gaussianity
should I look for it here or here?



Never mind that none of these data existed – and it would take decade(s) ...

Quasi-Linear Evolution of Compensated Cosmological Perturbations: The Nonlinear Sigma Model

Before matter
domination

Andrew H. Jaffe*

Enrico Fermi Institute and Department of Astronomy & Astrophysics

5640 S. Ellis Avenue, Chicago, IL 60637-1433

NASA/Fermilab Astrophysics Center, Fermi National Accelerator Laboratory,

Batavia, IL 60510-0500

Thus, the n -point spectra work out to be

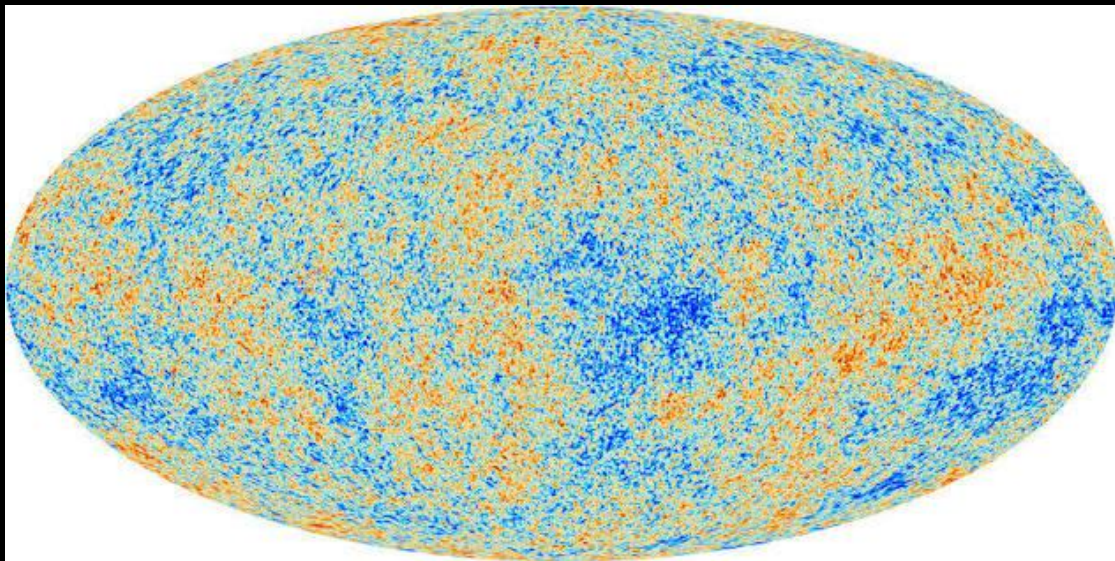
$$P_n = \left(\frac{G\phi_0^2}{10\sqrt{N}}\eta^2 \right)^n g_n \frac{k^{3-n}}{N^{n/2-1}}. \quad (56)$$

ments to dominate over higher-order effects? From Section II E above, the leading nonlinear contribution to the higher moments is

$$P_{n,\text{nl}}(k) = Q_n P_2^{n-1} = Q_n \left(\frac{G\phi_0^2}{10\sqrt{N}}\eta^2 \right)^{2(n-1)} (g_2 k)^{n-1}, \quad (61)$$

where this equation is actually only correct for $n = 3$ with equilateral triangles, where $Q_3 = 3Q$. (For $n > 3$, the right-hand side of this equation should be a sum over the

Answer



d'oh

We addressed the question of which of the CMB or LSS is better poised to detect primordial non-gaussianity of several varieties. We used the bispectrum as a discriminating statistic since it is the lowest-order quantity that has zero expectation value for a gaussian field. We considered three forms of non-gaussianity: one in which the gravitational potential contained a term that was the square of a gaussian field; another in which the density field was the square of a gaussian field; and a third that resembles that expected from topological defects. We showed that in all cases, the CMB is likely to provide a better probe of such non-gaussianity. One of these models produced a mass bispectrum that mimicked a scale-dependent non-linear bias, and the others mimicked a scale-independent non-linear bias.

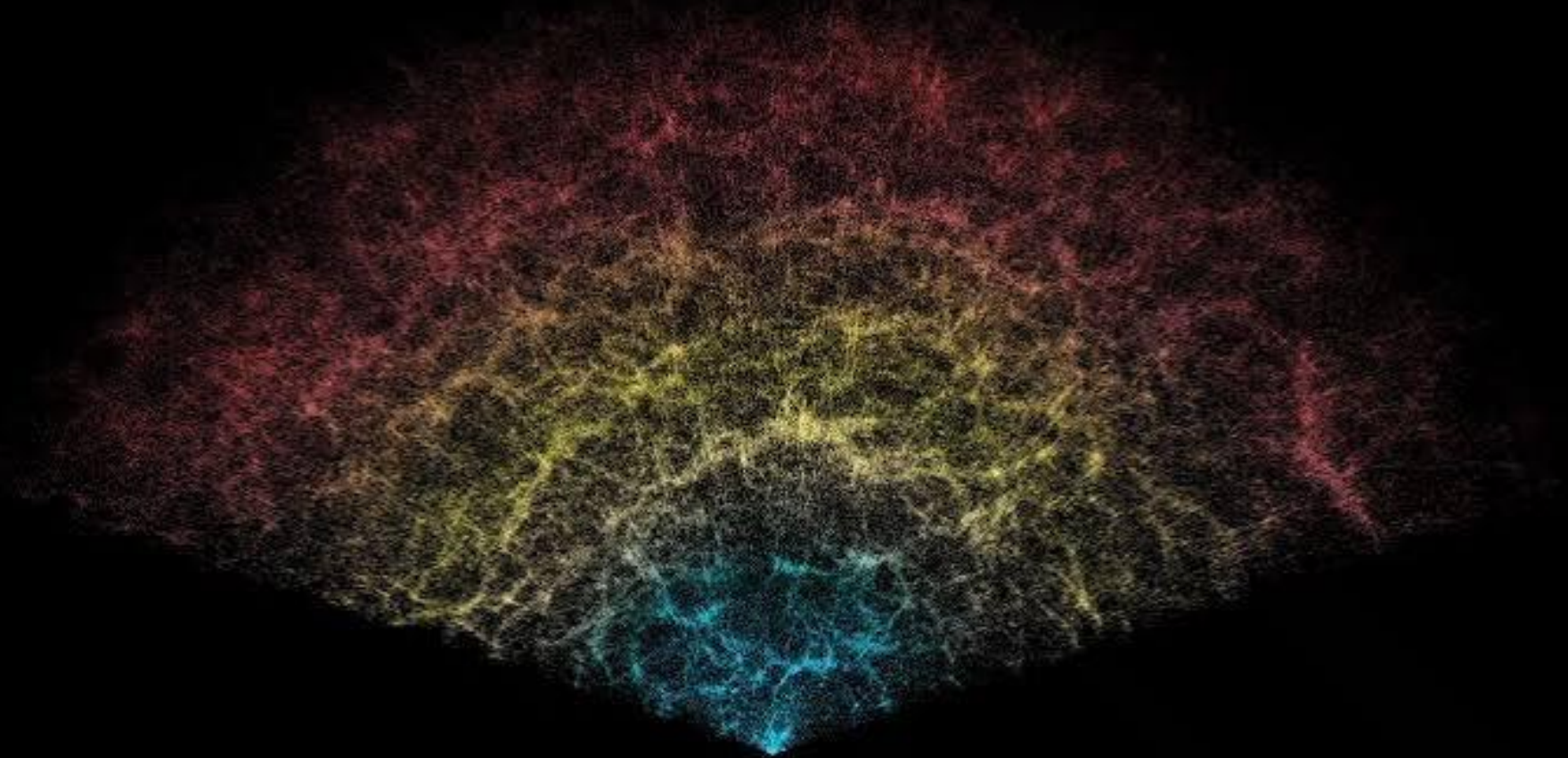
arXiv:astro-ph/9906301

Look
here!



makeameme.org

2dFGRS → SDSS → BOSS → DESI



Characterisation of the maps

Statistics

- **Power Spectrum $P(\mathbf{k})$** / 2-point correlation function. Gaussian part of the signal

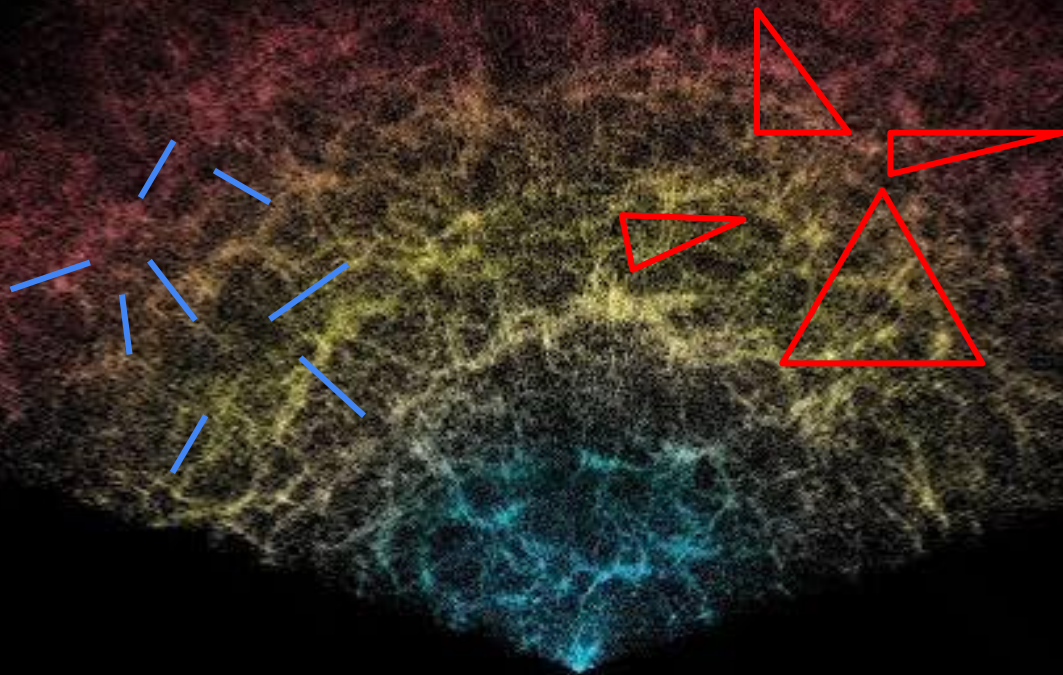
$$\langle \delta_{\mathbf{k}_1} \delta_{\mathbf{k}_2} \rangle = (2\pi)^3 \delta^D(\mathbf{k}_1 + \mathbf{k}_2) P(\mathbf{k}_1)$$

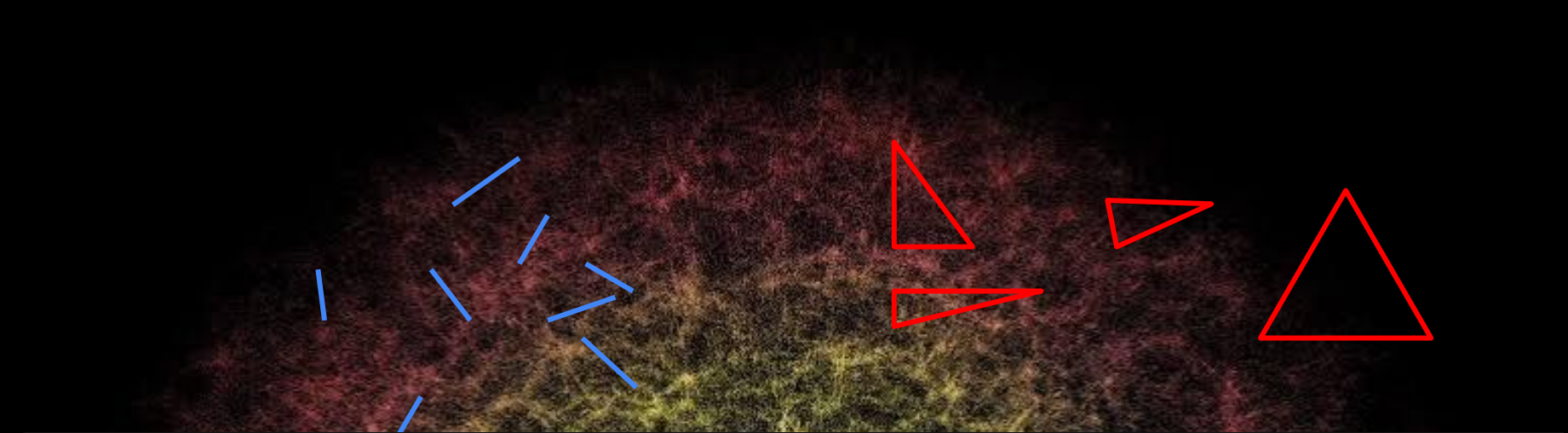
- **Bispectrum $B(\mathbf{k}_1, \mathbf{k}_2)$** / 3-point correlation function. Non-Gaussian part of the signal.

$$\langle \delta_{\mathbf{k}_1} \delta_{\mathbf{k}_2} \delta_{\mathbf{k}_3} \rangle = (2\pi)^3 \delta^D(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) B(\mathbf{k}_1, \mathbf{k}_2)$$

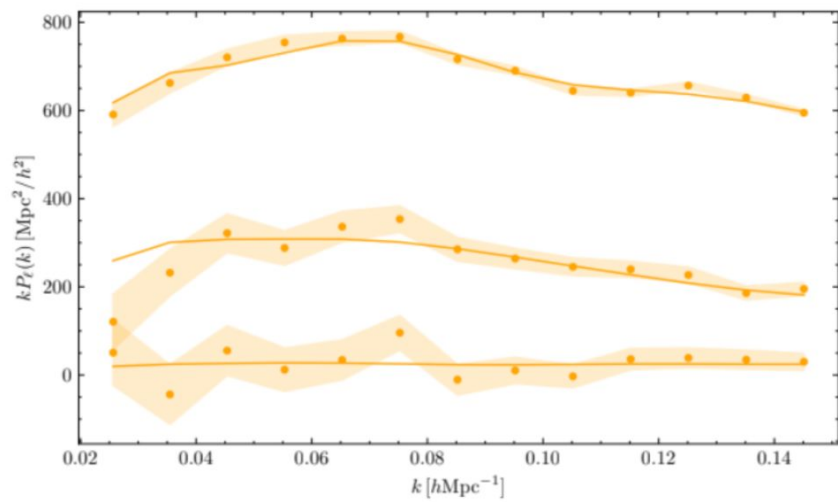
- In the early Universe (CMB): $B \neq 0 \iff f_{\text{nl}}, g_{\text{nl}} \neq 0$
- In the late Universe (galaxies) $B_{\text{gal}} = B_{\text{prim}}(f_{\text{nl}}, g_{\text{nl}}) \oplus B_{\text{grav}}(\Omega, \mathbf{b}_{\text{gal}})$

$$B_{\text{prim}}(f_{\text{nl}}, g_{\text{nl}}) \ll B_{\text{grav}}(\Omega, \mathbf{b}_{\text{gal}}) \longrightarrow B_{\text{prim}} \text{ not easy to detect!}$$

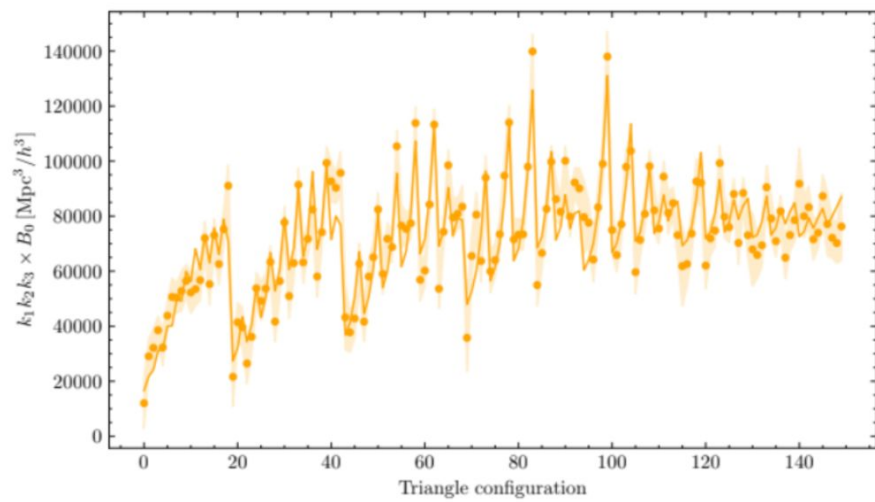




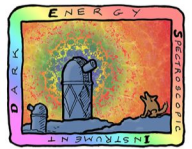
$$\langle \delta_{\mathbf{k}_1} \delta_{\mathbf{k}_2} \rangle = (2\pi)^3 P(\mathbf{k}) \delta^D(\mathbf{k}_1 + \mathbf{k}_2)$$



$$\langle \delta_{\mathbf{k}_1} \delta_{\mathbf{k}_2} \delta_{\mathbf{k}_3} \rangle = (2\pi)^3 B(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) \delta^D(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3)$$



Some decades later.....



DARK ENERGY
SPECTROSCOPIC
INSTRUMENT

U.S. Department of Energy Office of Science

Dark Energy Spectroscopic Instrument

Five target classes

50 million redshifts
in 5 years

DESI (2021-2026)

3 million QSOs

Lya $z > 2.1$

Tracers $0.9 < z < 2.1$

16 million ELGs

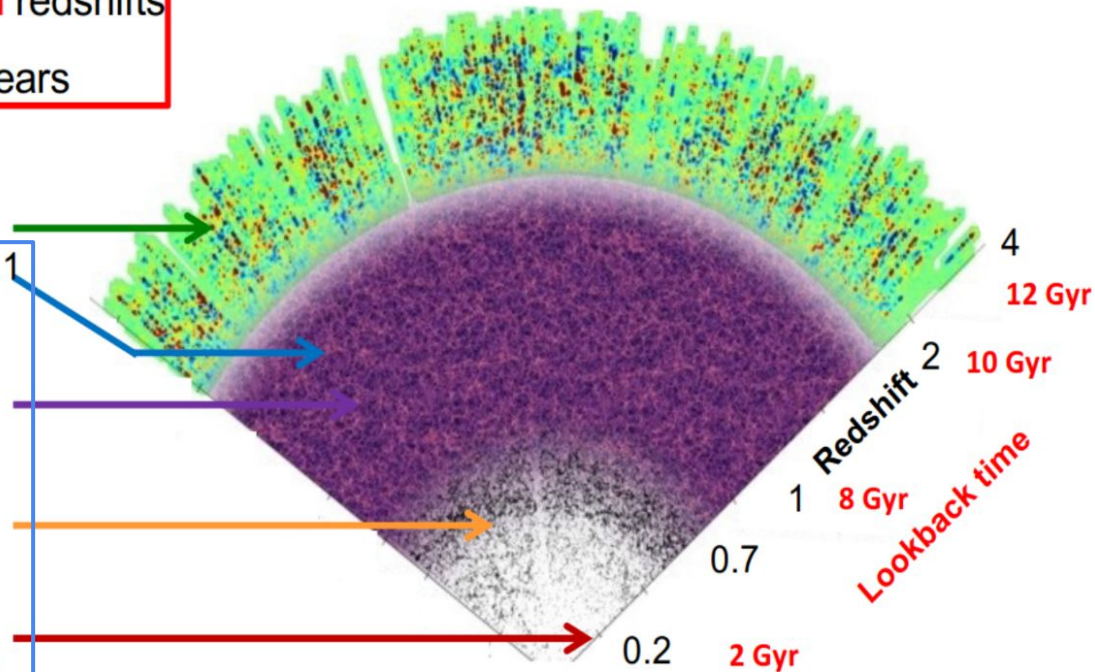
$0.6 < z < 1.6$

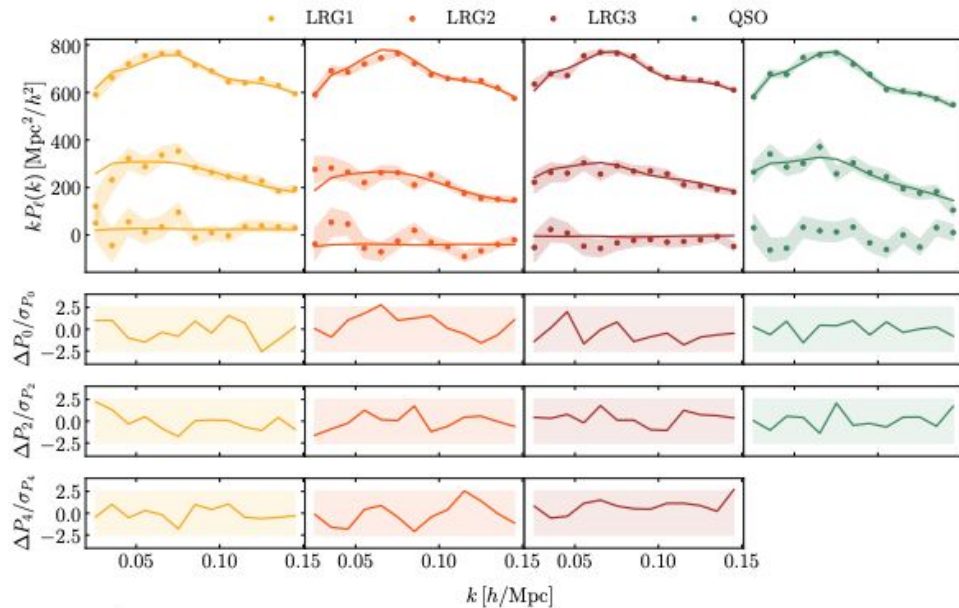
8 million LRGs

$0.4 < z < 1.0$

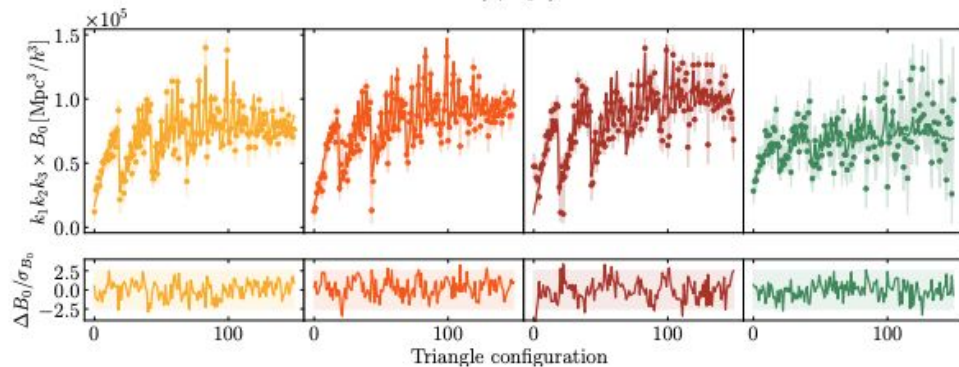
**13.5 million
Brightest galaxies**

$0.0 < z < 0.4$





Measurement of DESI data



Looks ugly (and that's only the monopole)

Bispectrum: challenges and limitations

- It is challenging both theoretically and computationally
- It is expensive to compute and evaluate
- It is complex to model beyond linear scales: can try tree level, loops in PT
- Or trade-off: fitting formula based on simulations
- In redshift space you get three quadrupoles and thinking of hexadecapoles gives you headache
- The size of the data vector makes estimating the covariance also a challenge
- The effect of sky cut (window) is very hard to model.

Can't be purist Done is the enemy of perfect: work with good enough

The heroes



Hector Gil-Marin



Adriana Nadal-Matosas



Sergi Novell-Masot



Paula Royo i
Salvaterra



Davide
Gualdi

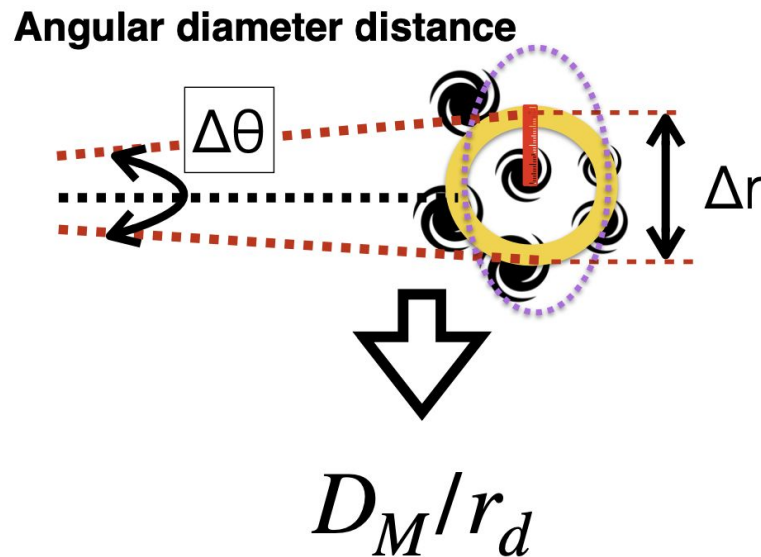
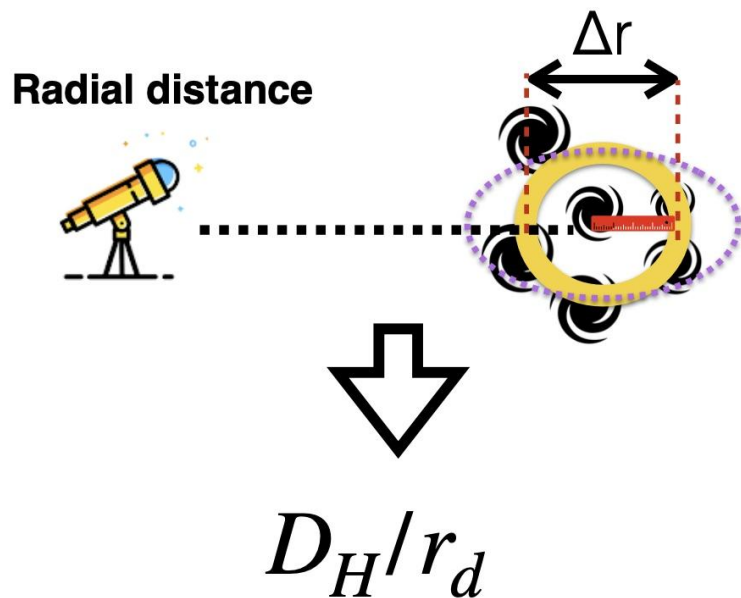


Emanuele Fondi



Leonid Saredine

distances



Relative to a fiducial:

$$\alpha_{\parallel} = [D_H/r_d]/[D_H/r_d]^{\text{fid}}$$

$$\alpha_{\perp} = [D_M/r_d]/[D_M/r_d]^{\text{fid}}$$

Fig. credit: H. Gil-Marin

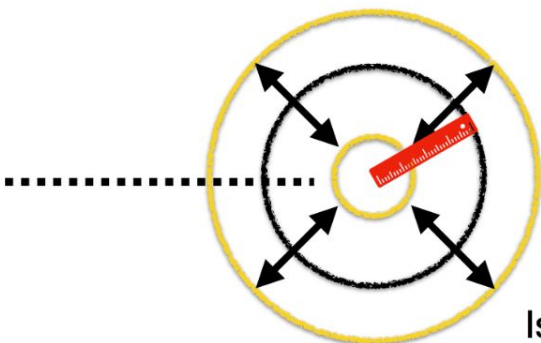


DARK ENERGY
SPECTROSCOPIC
INSTRUMENT

U.S. Department of Energy Office of Science

The BAO as a standard ruler

The along and across LOS BAO distortions can be reparametrized in



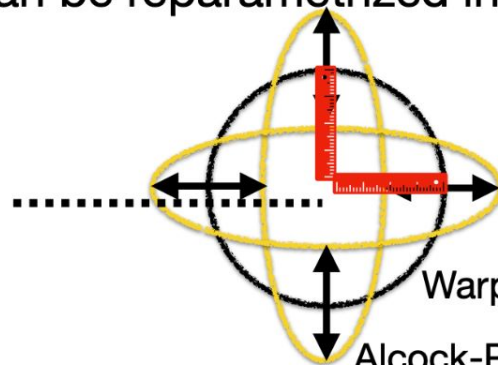
Isotropic re-scaling



$$\{[D_M(z)/r_d]^2 D_H(z)/r_d\}^{1/3} \equiv D_V(z)/r_d$$

For aficionados →

$$[\alpha_{\parallel} \alpha_{\perp}^2]^{1/3} \equiv \alpha_{\text{iso}}$$



Warp of scales

Alcock-Paczynski effect



$$D_M(z)/D_H(z)$$

$$\alpha_{\perp}/\alpha_{\parallel} \equiv \alpha_{AP}$$

Alcock-Paczynski
parameter

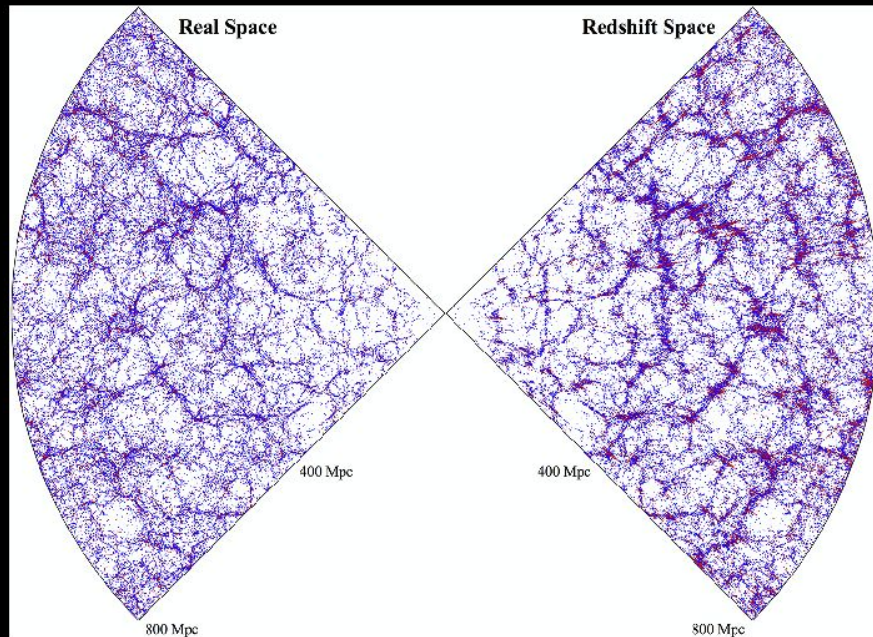
Fig. credit: H. Gil-Marin

Redshift space distortions and growth of structure

Peculiar velocities (deviations from the Hubble flow)

Distort the galaxy map along the LOS 🙄

Are sourced by gravity and total matter 😊



In the linear regime

$$f = \frac{d \ln \delta}{d \ln a} \approx \Omega_m^{\gamma}$$

Fig. Courtesy of ICC Durham

ShapeFit

Brieden et al., arXiv: 2106.11931,

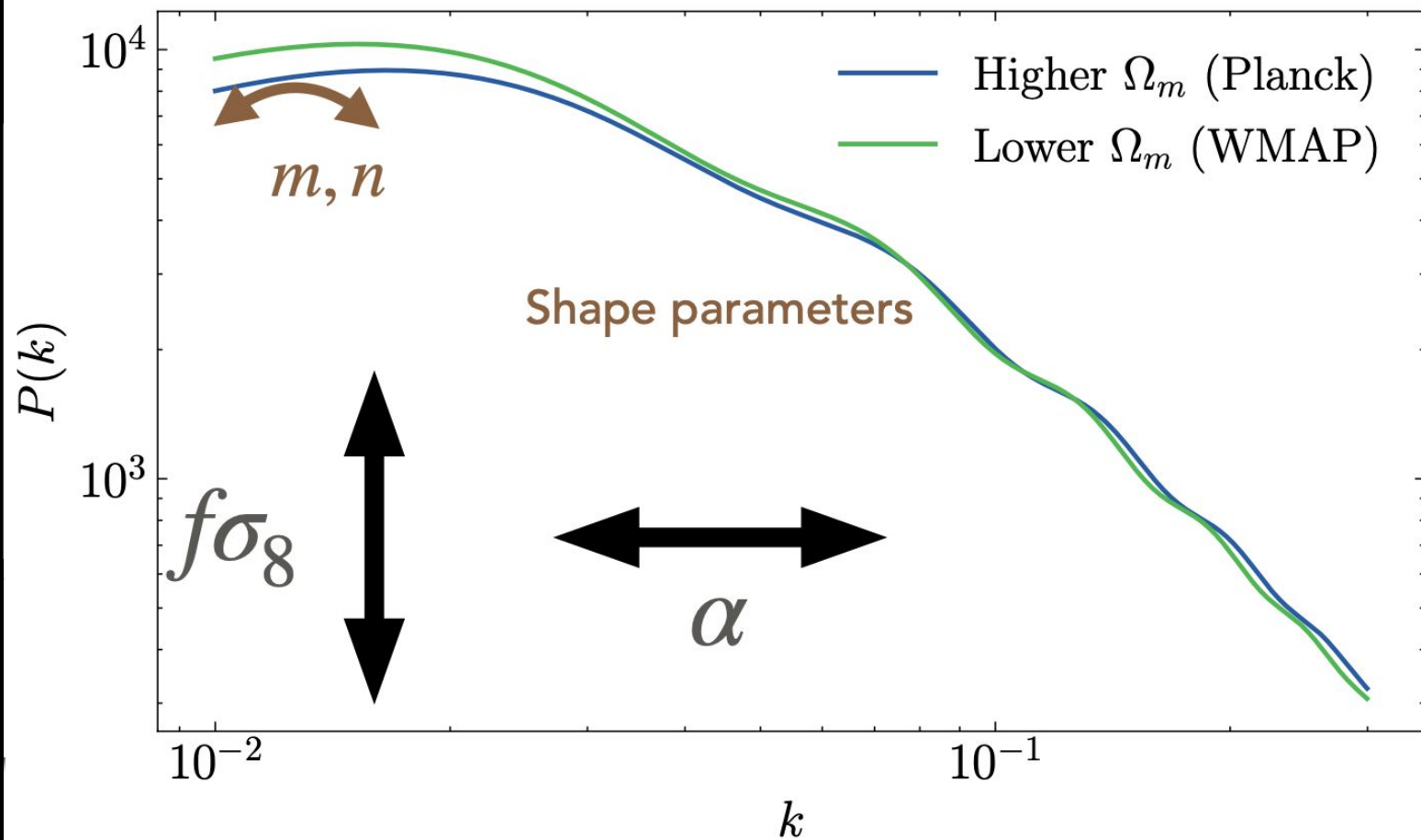
Brieden et al arXiv:2106.07641

$$m = \frac{d}{dk} \left(\ln \left[\frac{P_{\text{no-wiggle}}^{\text{lin}} \left(\frac{k_p}{s}, \boldsymbol{\Omega} \right) / \mathcal{P}_{\mathcal{R}} \left(\frac{k_p}{s}, \boldsymbol{\Omega} \right)}{P_{\text{no-wiggle}}^{\text{lin}} (k_p, \boldsymbol{\Omega}^{\text{ref}}) / \mathcal{P}_{\mathcal{R}} (k_p, \boldsymbol{\Omega}^{\text{ref}})} \right] \right) \Big|_{k=k_p}$$

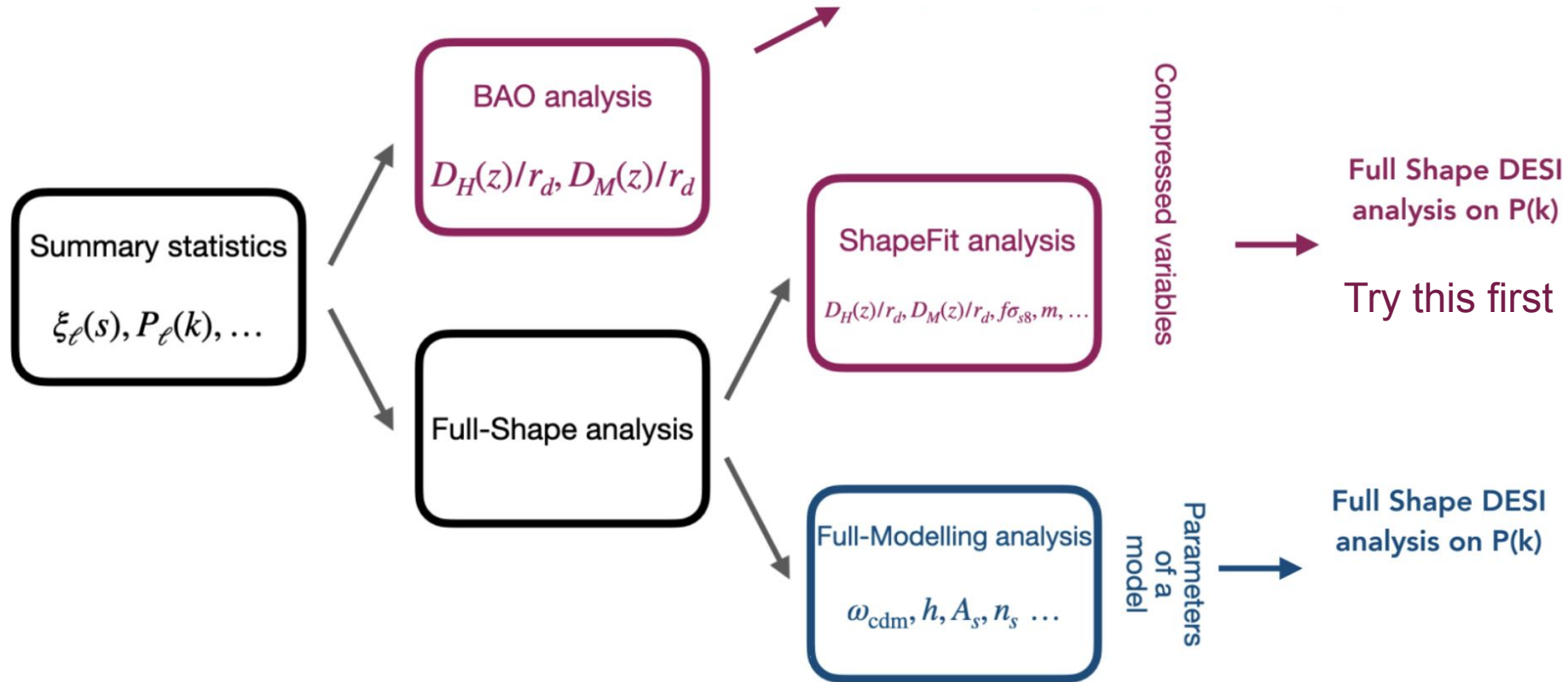
$$m = \frac{d}{dk} \left(\ln \left[\frac{\left(\frac{r_d^{\text{fid}}}{r_d} \right)^3 \cdot T_{\text{nw}}^2 \left(\left(\frac{r_d^{\text{fid}}}{r_d} \right) \cdot k \right)}{T_{\text{nw}}^{\text{fid}^2} (k)} \right] \right) \Big|_{k=k_p}$$

One extra effective parameter to the “usual” compressed variables (physical parameters).

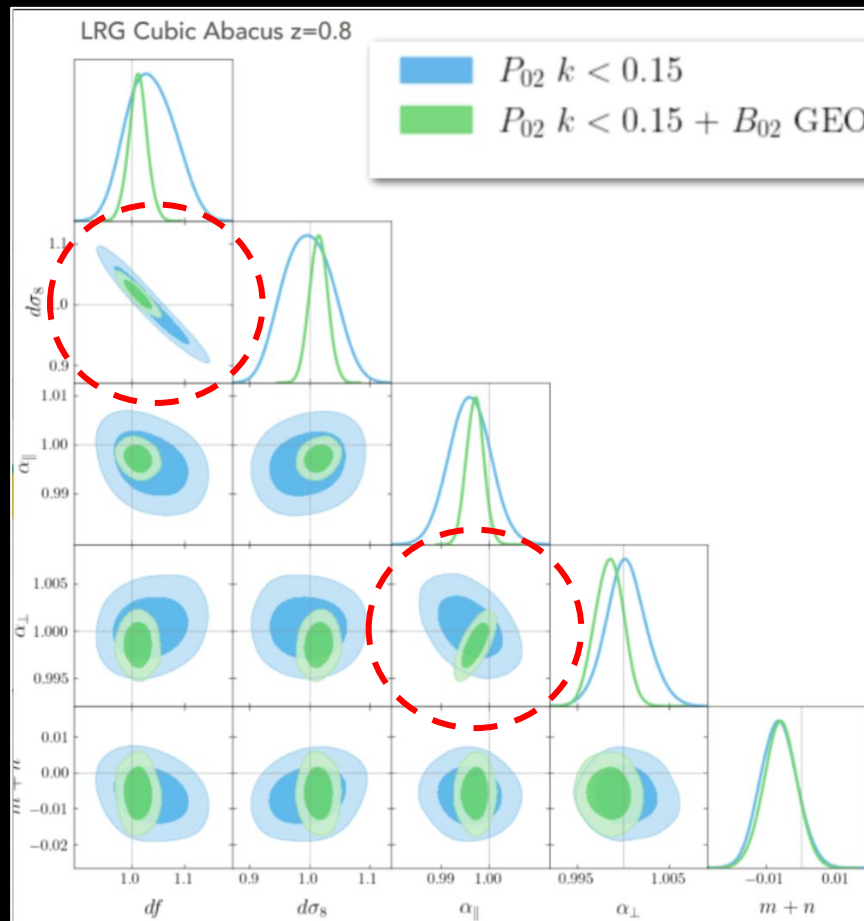
Quite generic: mops up things like fnl, neutrino masses, dark radiation, any expansion history, generic growth histories etc.



COSMOLOGICAL SUMMARY STATISTICS



WHY GOING THROUGH THE PAIN?



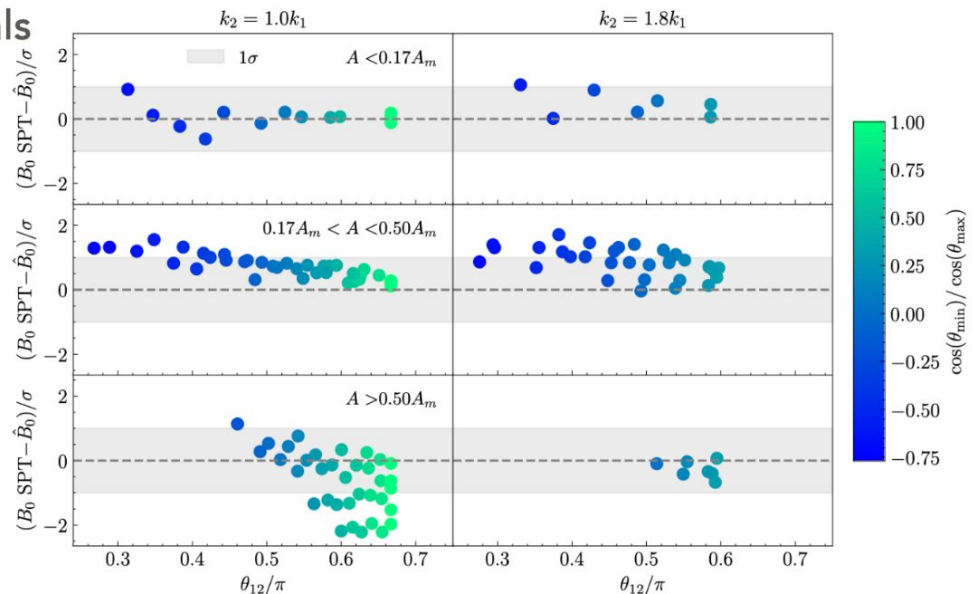
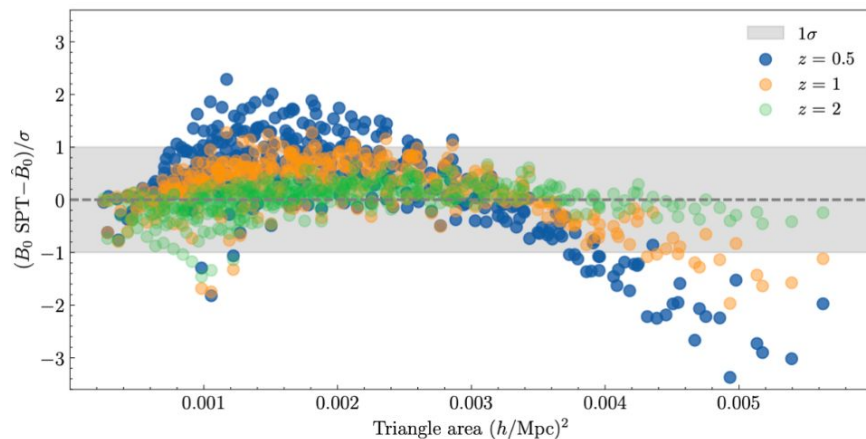
Combining power spectrum and bispectrum multipoles removes crucial degeneracies between parameters:

- $f - \sigma_8$ degeneracy
- $\alpha_{||} - \alpha_{\perp}$

- It is challenging both theoretically and computationally
- It is expensive to compute and evaluate
- It is complex to model beyond linear scales: can try tree level, loops in PT
- **Or trade-off: fitting formula based on simulations**
 - ➡ **GEO-FPT (Novell-Masot et al 2023)**
- In redshift space you get three quadrupoles and thinking of hexadecapoles gives you headache
 - ➡ With two quadrupoles one maximizes information gain (Gualdi et al 2022)
- The size of the data vector makes estimating the covariance also a challenge
 - ➡ Could calibrate a semi-analytical matrix
- The effect of sky cut (window) is very hard to model.
Still unsolved for multipoles.

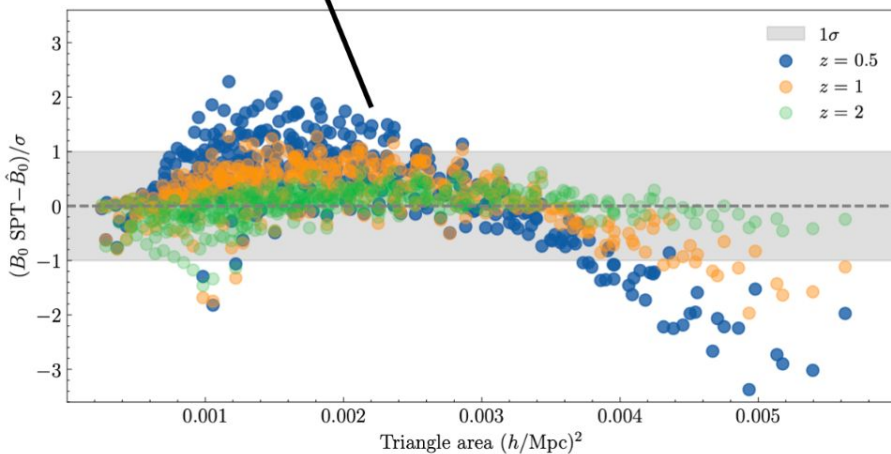
MODELLING THE BISPECTRUM

- Tree-level bispectrum, only valid up to scales of $k \sim 0.08h\text{Mpc}^{-1}$
- $B^{\text{SPT}}(\mathbf{k}_1, \mathbf{k}_2) = D_{\text{FoG}}^B(\mathbf{k}_1, \mathbf{k}_2) [2Z_1(\mathbf{k}_1)Z_1(\mathbf{k}_2)Z_2(\mathbf{k}_1, \mathbf{k}_2)P^L(k_1)P^L(k_2) + 2\text{perm.}]$
- How does it “fail”? Look at the residuals

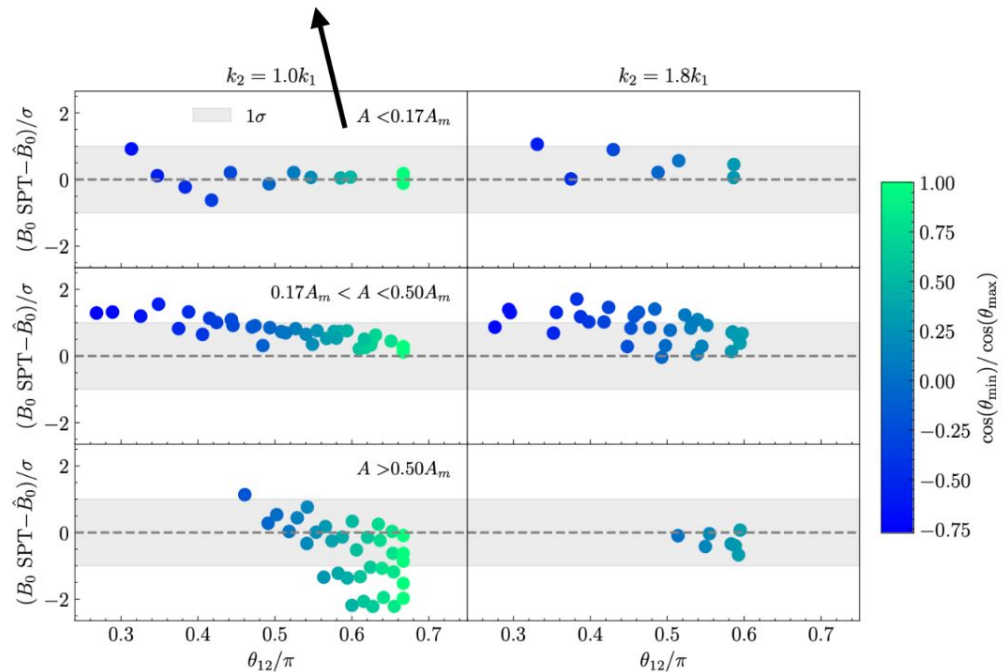


MODELLING THE BISPECTRUM

(Parabolic) dependence on
the area of the triangle $\rightarrow f_4 A + f_5 A^2$



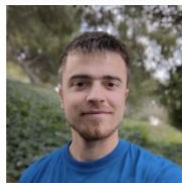
At fixed ratio k_2/k_1 , dependence on θ_{12} ,
dependence on "flatness" or "squeezedness" $\rightarrow f_2 \frac{\cos \theta_{med}}{\cos \theta_{max}} + f_3 \frac{\cos \theta_{min}}{\cos \theta_{max}}$



MODELLING THE BISPECTRUM

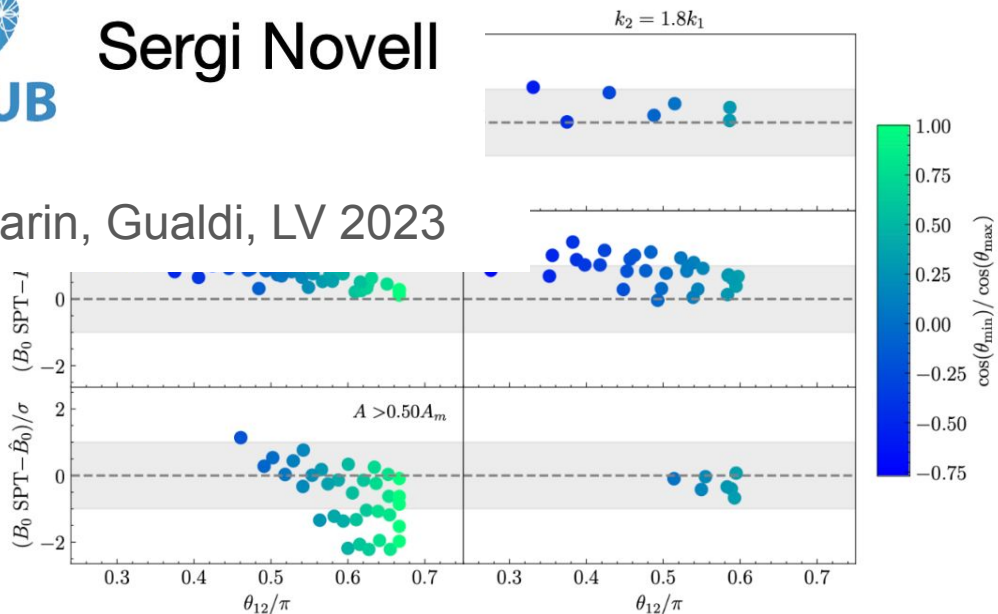
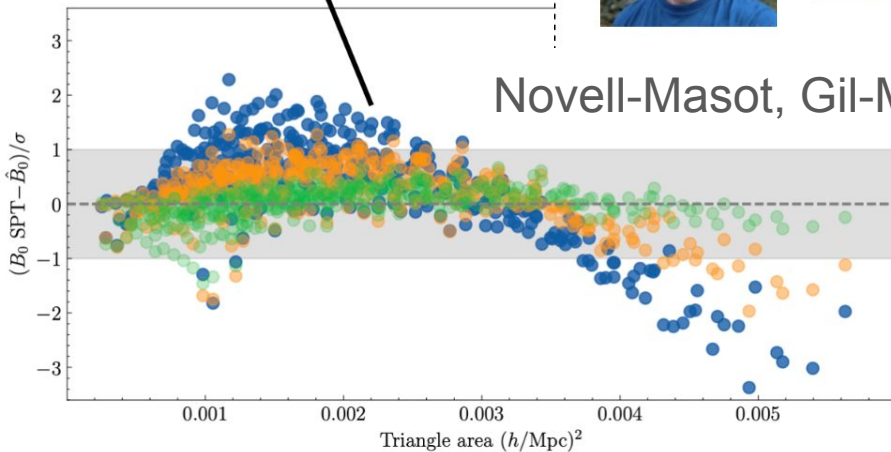
At fixed ratio k_2/k_1 , dependence on θ_{12} ,
dependence on "flatness" or "squeezedness" $\rightarrow f_2 \frac{\cos \theta_{med}}{\cos \theta_{max}} + f_3 \frac{\cos \theta_{min}}{\cos \theta_{max}}$

(Parabolic) dependence on
the area of the triangle $\rightarrow f_4 A + f_5$

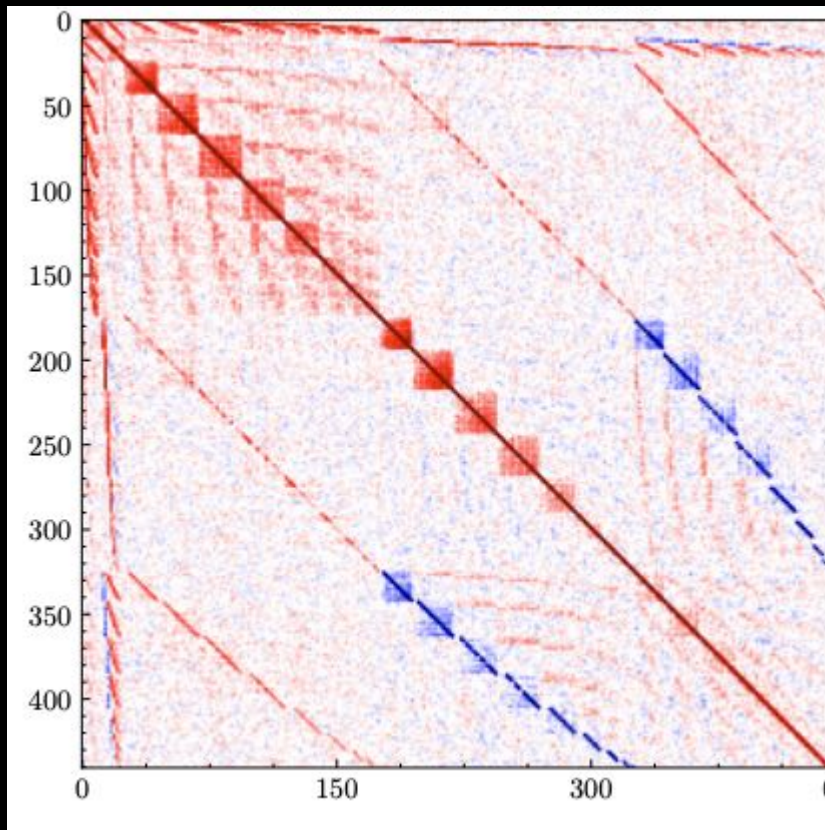


Sergi Novell

Novell-Masot, Gil-Marin, Gualdi, LV 2023



Covariance matrix



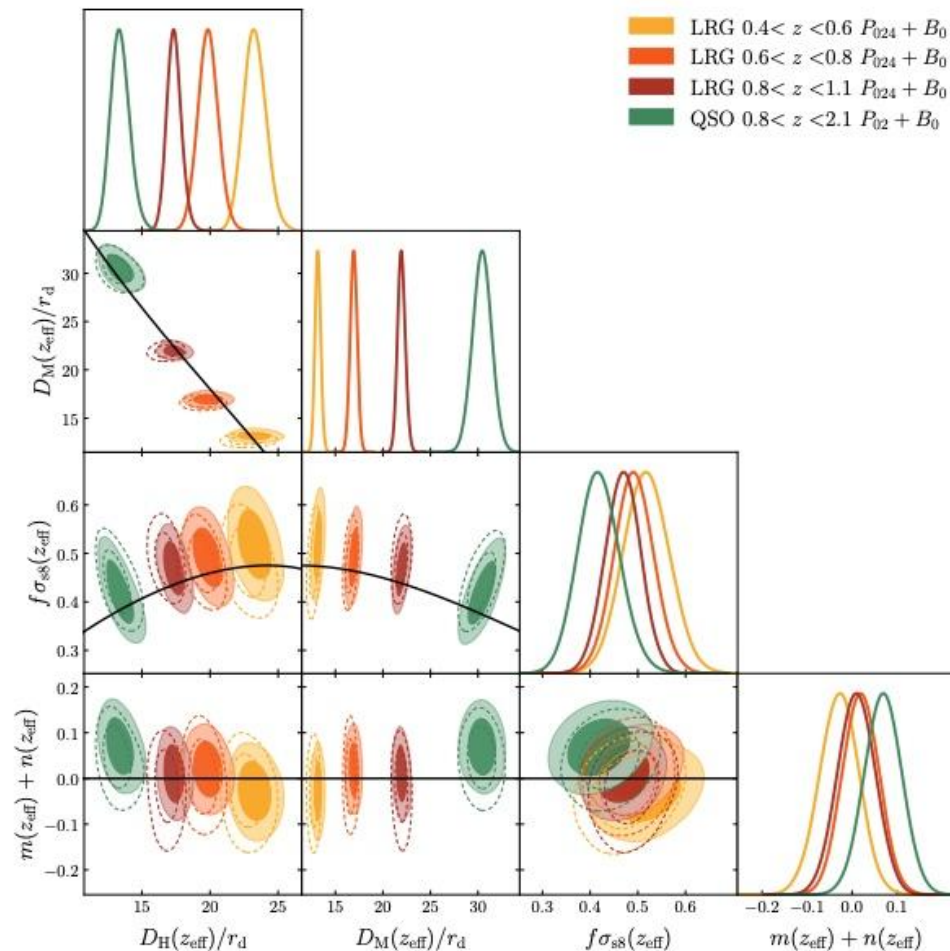
The number of simulations needed increases

Efforts to compute it analytically and calibrate

Gualdi et al 2020. Novell et al 2023, Royo i Salvaterra in prep.



DESI DR1



* **LRG Data-vector:** P_{024} ($k < 0.15$) + B_0 ($k < 0.12$)

• **For QSO:** P_{02} ($k < 0.15$) + B_0 ($k < 0.12$)

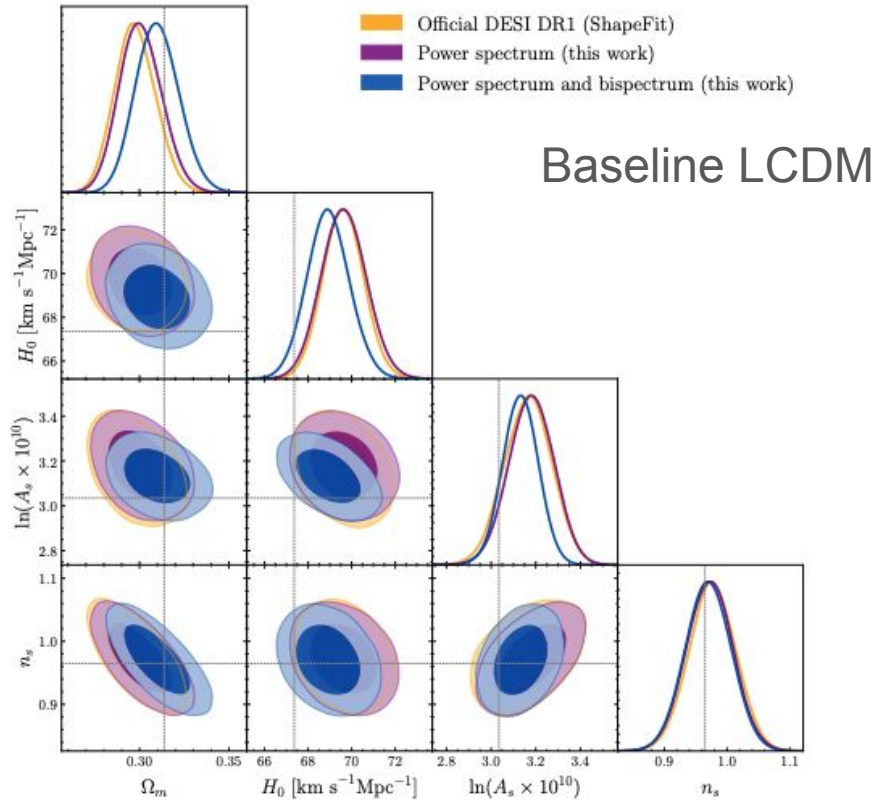
• Error-bars including **statistical** and **systematic** error budgets.

• We **constrain** (at 10%) **and** (at 8%), and obtain **compatible** (and **improved in some tracers**) constraints to official DESI power spectrum analysis.



Novell-Masot (DESI collab) 2025

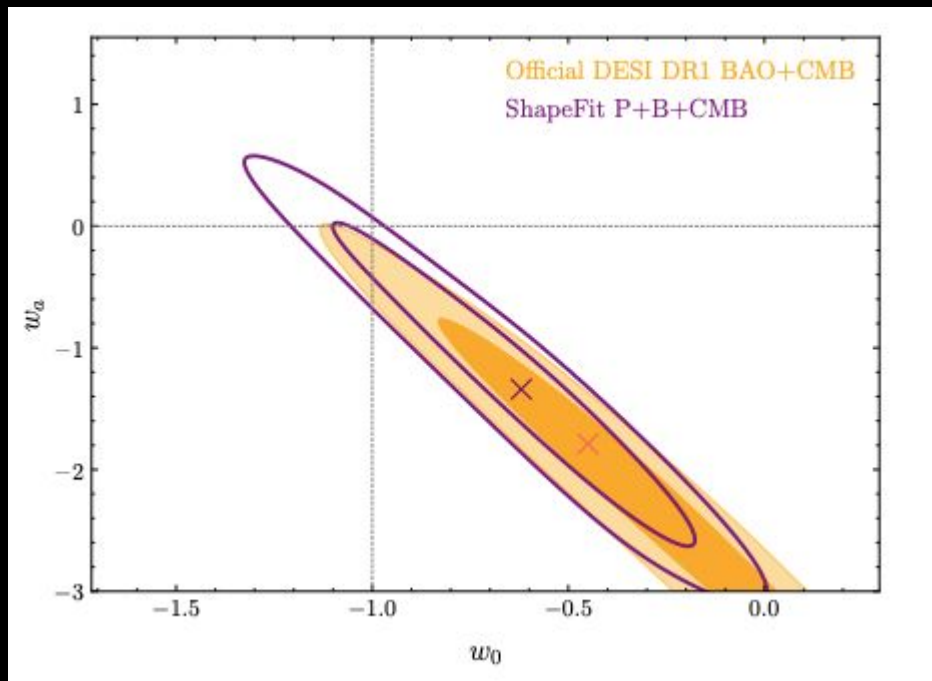
What does it mean for cosmology?



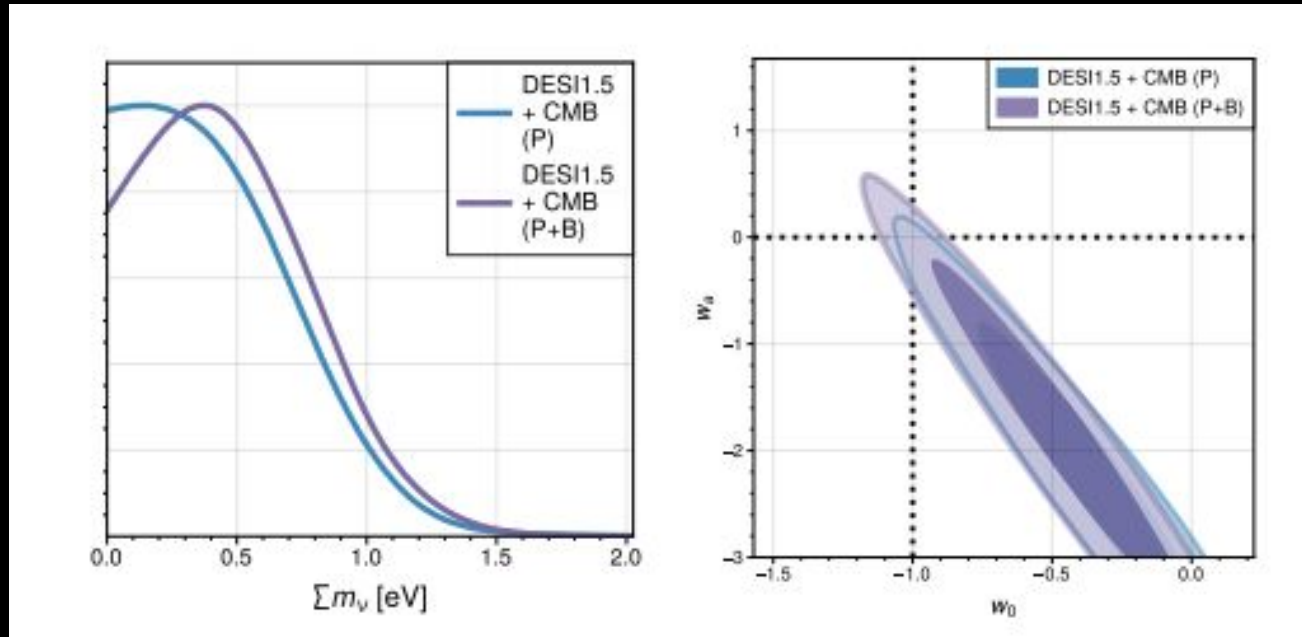
Novell-Masot+DESI 2026

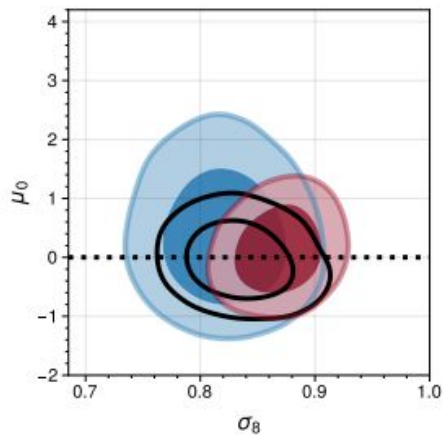
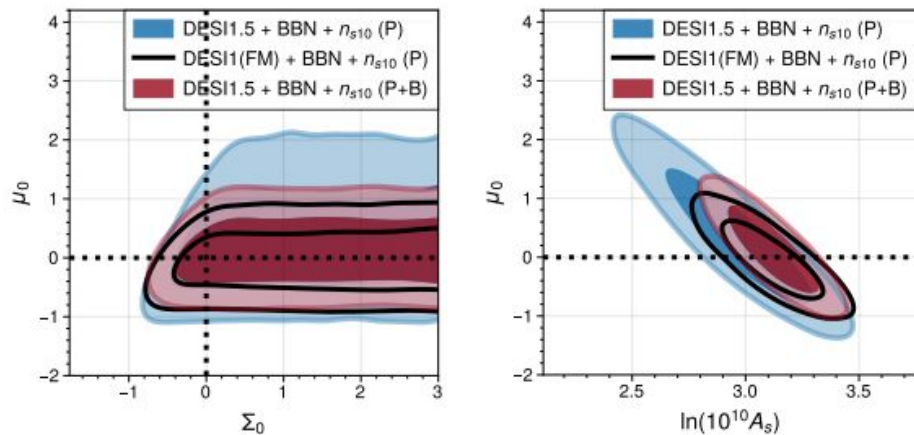


Is dark energy evolving?



In combination with BAO DR2





“Modification of gravity”

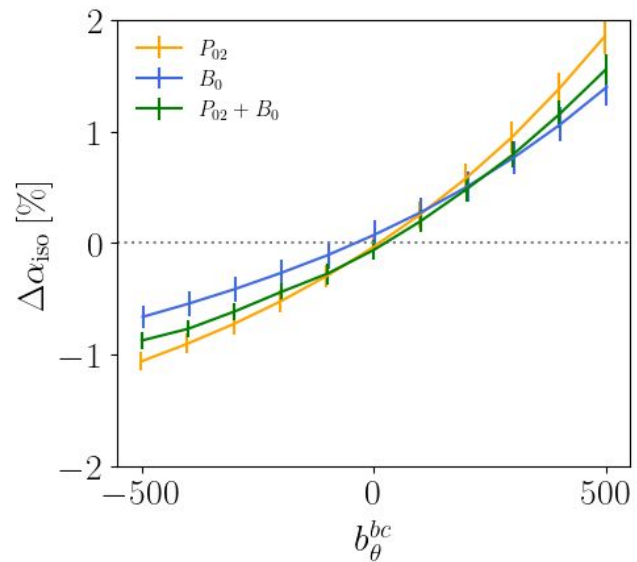
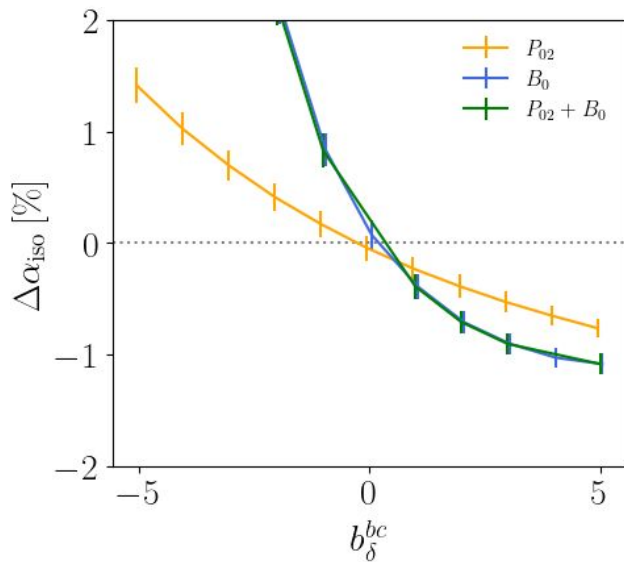
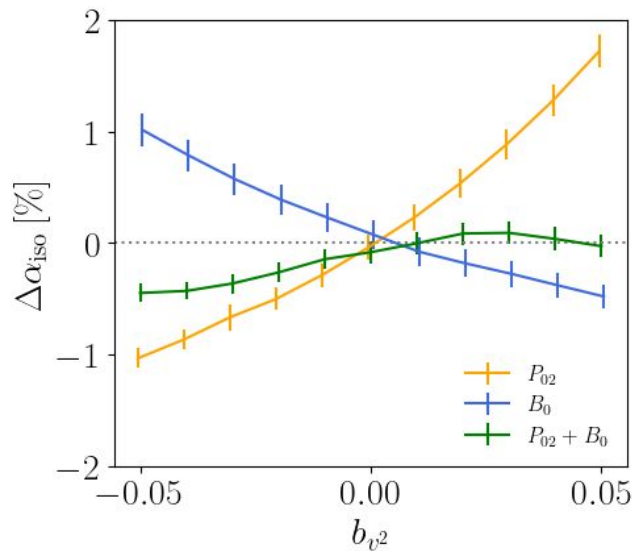
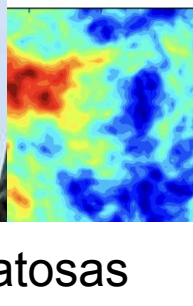
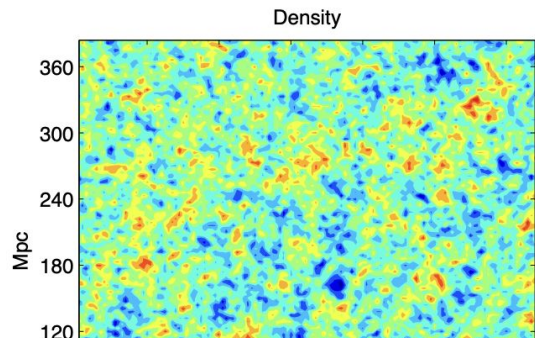
Forero-Sanchez et al 2026

Biases in BAO $P(k)$ e.g.,



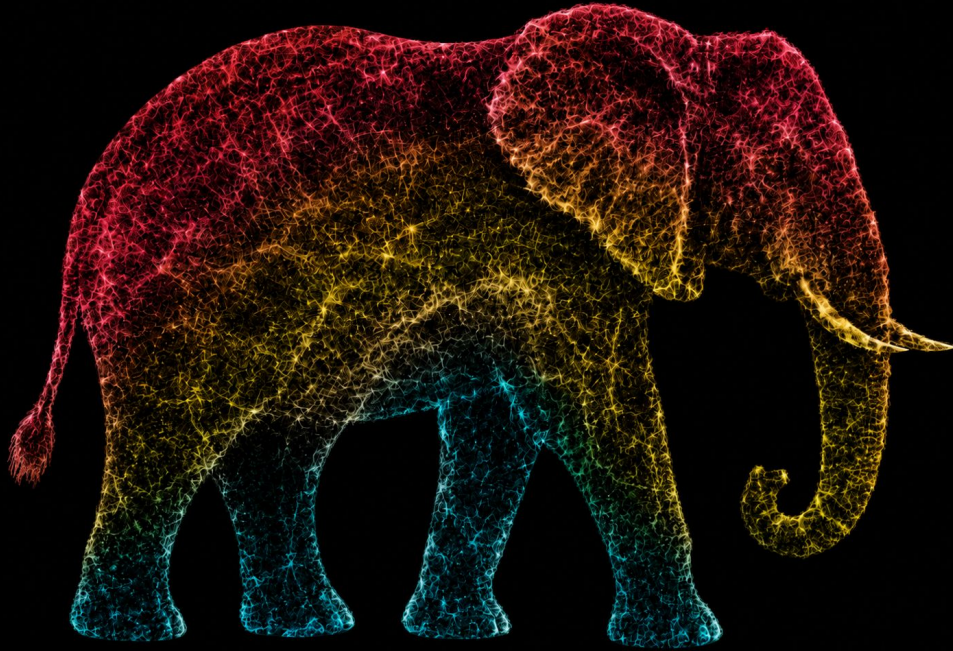
Adriana Nadal-Matosas

Unpublished

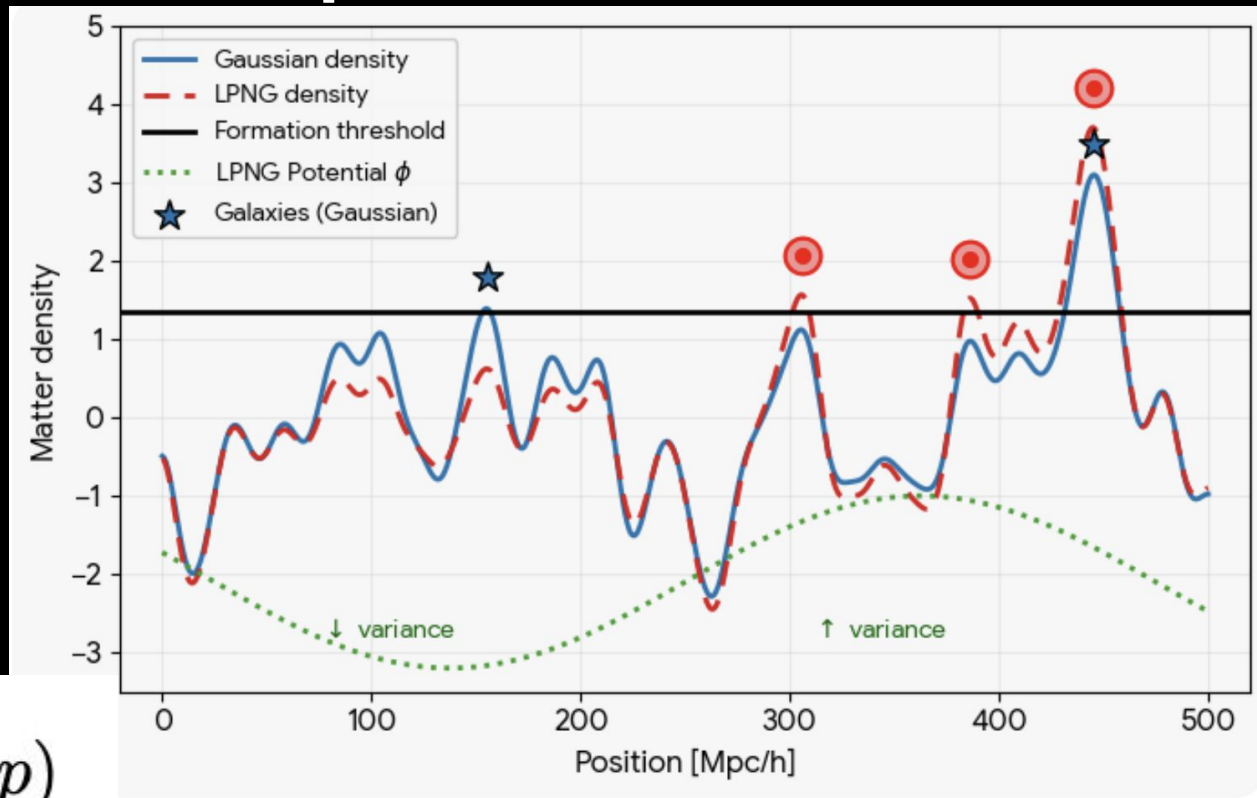


More to come with DR2 data

The anisotropic signal is key but the window is



What about primordial NG?



$$b_{\phi}^s = 2\delta_c(b_1 - p)$$

Now LSS is back in business for PNG (local)

How do you simulate?

What about assembly bias?

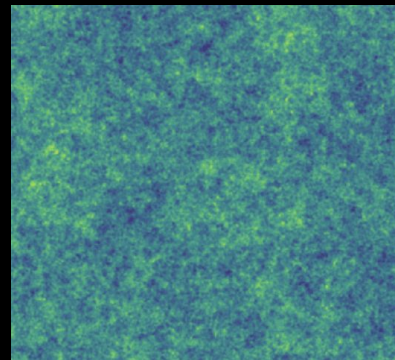


GENGARS code



Emanuele Fondi

$B(k_1, k_2, k_3)$
& $P(k)$



Now LSS is back in business for PNG (local)

How do you simulate?

What about assembly bias?

Extended press Schechter does it...

$$p = 1 - \frac{1}{\delta_c} \left[\frac{\omega_f e^{-\omega_f^2/2}}{\sqrt{2\pi} \operatorname{erfc}\left(\frac{\omega_f}{\sqrt{2}}\right)} - 1 \right]$$



Emanuele Fondi

$$b_{\phi}^s = 2\delta_c(b_1 - p)$$

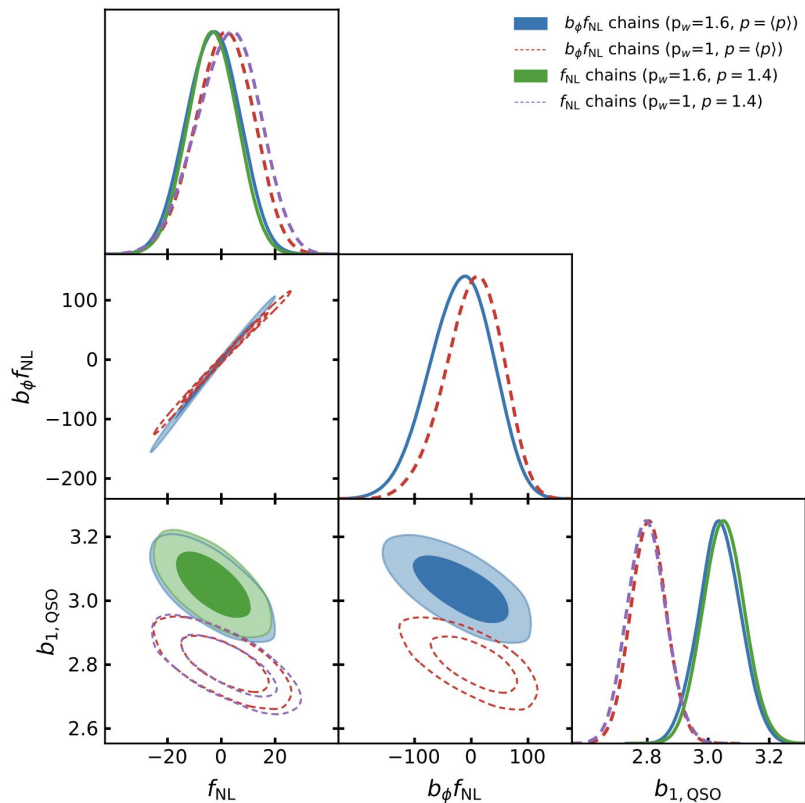
$$\omega_f \equiv \frac{\delta_c \left(\frac{1}{D(z_f)} - \frac{1}{D(z_o)} \right)}{\sqrt{\sigma^2(fM_o) - \sigma^2(M_o)}},$$

Property of the host halo (ie. dark matter)
related to its formation time does not know about NG

For LRG and QSO can easily be calibrated from
observational properties (e.g., colors)

Fondi et al 2024

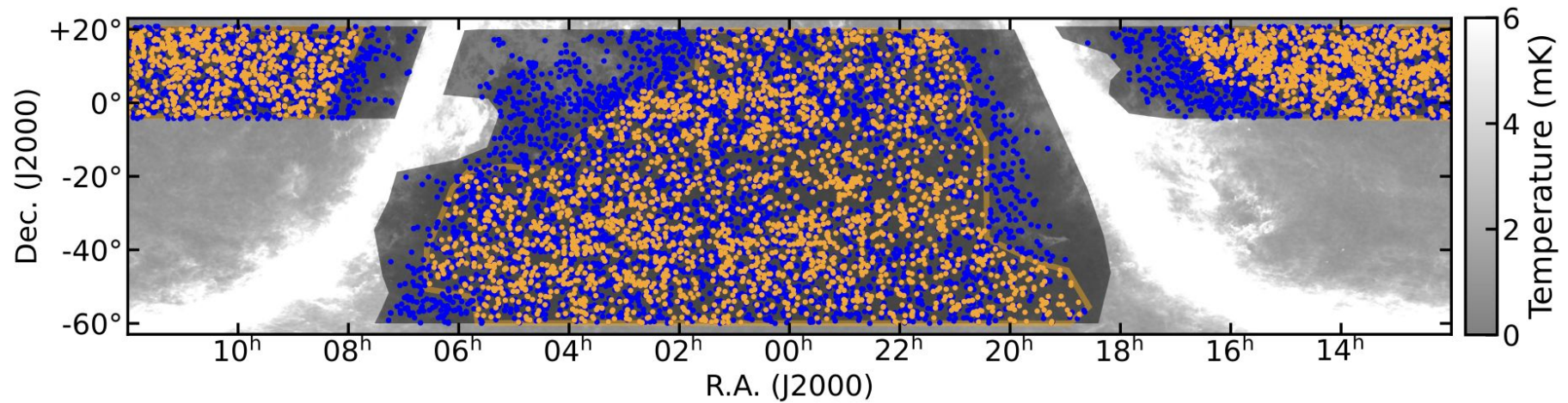
DESI Quasars: Non-Gaussianity from halo bias



$$f_{\text{NL}} = -3.3 \pm 9.2$$

Chaussidon et al 2024
+ Fondi et al 2025

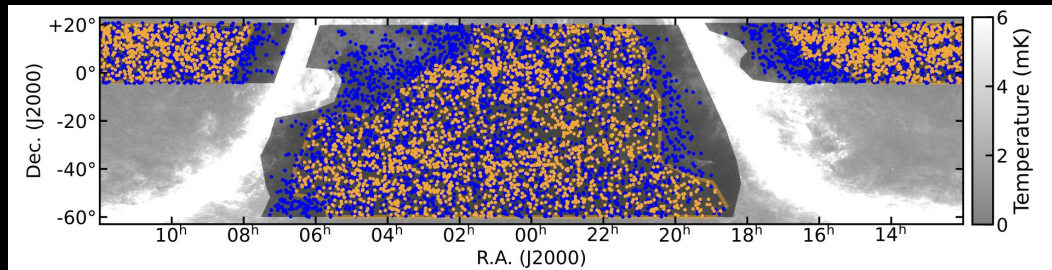
Primordial NG affects the halo abundance and the mass function



Aguena et al 2025

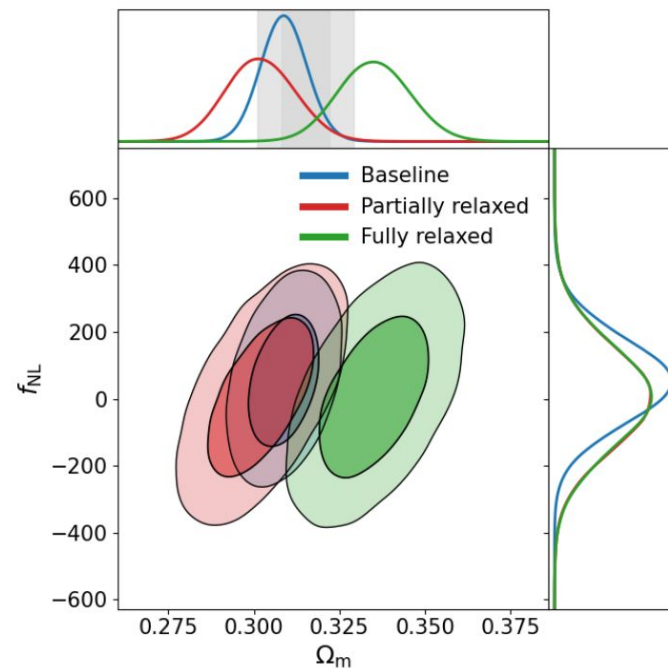
ACT DR6 SZ clusters

Primordial NG affects the halo abundance and the mass function



Leonid Sarieddine

$$f_{\text{NL}} = 55.3 \pm 125$$



Sarieddine et al 2026

I think I have made it clear who took the sensible decision

following the first commandment of cosmology:
“Thou shall use the CMB”

The late Universe is interesting but complicated and messy

it does not give up its secrets quite so gracefully.

